Jeffrey Utley

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RESEARCH SUMMARY

PhD candidate at Purdue University developing statistically validated synthetic data generation algorithms and reproducible scientific software for turbulence-driven imaging and simulation. Three years of DoD-funded research in active collaboration with AFRL and AFIT, with first-author publications in *Journal of the Optical Society of America A*, *Optical Engineering*, and SPIE proceedings, and pip-installable Python packages validated on measured datasets. Targeting Research Scientist and R&D roles at national laboratories and in industry, with focus on modeling, estimation, and simulation; computational imaging; signal processing; and scientific software.

TECHNICAL SKILLS

Methods: synthetic data generation; inverse problems and parameter estimation; spectral analysis (FFT, PSD, PCA); numerical optimization (first-order methods, Mirror Descent); statistical modeling (Gaussian models, Poisson point processes); Bayesian inference; signal and image processing
Programming: Python, C/C++, MATLAB, Julia
Scientific Computing: NumPy, SciPy, pandas, Matplotlib, Jupyter, OpenCV, seaborn
Machine Learning: PyTorch, JAX
HPC & Systems: Linux/UNIX, bash, SSH, Slurm (Purdue clusters)
Reproducibility & Tools: Git/GitHub, pip/conda, setuptools/pyproject

EDUCATION

Ph.D., Mathematics — Purdue University Expected Dec 2026
Concentration: Computational Science • GPA: 4.00 • Advisors: Prof. Gregery Buzzard and Prof. Charles Bouman
Dissertation: Synthetic Data Generation for Aero-Optics
Selected Coursework: Foundations of Computational Imaging; Digital Image Processing; Digital Cameras & Fourier Optics; Deep Learning; Numerical Optimization; Numerical Linear Algebra; Numerical Analysis; Probability

B.S., Mathematics (Honors) — University of Tennessee, Knoxville May 2022
Minor: Computer Science • GPA: 3.98

RESEARCH EXPERIENCE

Graduate Research Assistant — Purdue University, Department of Mathematics Jan 2023 – Present

- Designed **ReVAR**, a novel data-driven algorithm producing synthetic data that matches measured statistics from physical experiments with worst-case **4% NRMSE** on the temporal power spectrum — a substantial improvement over existing methods.
- Developed **BoilingFlow**, extending a Fourier-based data synthesis method with novel automated parameter estimation and a spatial-anisotropy extension; validated on measured and simulated datasets.
- Delivered all results as modular, version-controlled, **pip-installable Python packages** with reproducible examples and full documentation; produced first-author publications in *Journal of the Optical Society of America A*, *Optical Engineering*, and SPIE proceedings.

AFRL Scholars Program (USRA) — Air Force Research Laboratory (virtual) May 2026 – Sept 2026 (upcoming)
Mentor: Jeremy Vorenberg

- Second consecutive AFRL Scholars selection; continues the Purdue–AFRL/AFIT collaboration on synthetic data generation for turbulence-driven imaging and simulation with Prof. Matthew Kemnetz.

Doctoral Research Fellow (ORISE) — Air Force Institute of Technology (virtual) May 2025 – Aug 2025

Mentor: Prof. Matthew Kemnetz

- Advanced the BoilingFlow algorithm as part of the Purdue–AFRL/AFIT collaboration; delivered weekly virtual technical briefings to AFIT faculty and contributed to a SPIE conference paper deliverable.

AFRL Scholars Program (USRA) — Air Force Research Laboratory May 2024 – Aug 2024

Mentor: Prof. Matthew Kemnetz

- Developed and evaluated the ReVAR algorithm against measured wind tunnel data; delivered an end-of-summer technical briefing to AFRL staff and contributed to a SPIE conference paper.

RESEARCH SOFTWARE

AOModel — pip-installable Python package implementing the ReVAR algorithm for data-driven synthesis of optical turbulence data. Validated on two wind tunnel datasets with worst-case 4% NRMSE on the temporal power spectrum. Includes thorough documentation and reproducible examples.
github.com/jeffreyutley/aomodel_public

BoilingFlow — pip-installable Python package implementing the BoilingFlow algorithm, extending a Fourier-based data synthesis method with automated parameter estimation and a spatial-anisotropy extension. Validated on two wind tunnel datasets. Includes thorough documentation and reproducible examples.
github.com/jeffreyutley/boiling_flow

PUBLICATIONS

Refereed Journal Articles

Utley, J., Buzzard, G., Bouman, C., & Kemnetz, M. ReVAR: A data-driven algorithm for generating aero-optic phase screens. *Journal of the Optical Society of America A* **(in press).** Preprint: arXiv:2604.02326.

Utley, J., Buzzard, G., Bouman, C., & Kemnetz, M. Boiling flow estimation for aero-optic phase screen generation. *Optical Engineering* **(in press)**. Preprint: arXiv:2601.12171.

Lind, J. & **Utley, J.** (2022). Phase transition for a family of complex-driven Loewner hulls. *Involve, a Journal of Mathematics*, 15(3), 447–474. DOI: 10.2140/involve.2022.15.447. Preprint: arxiv:2106.14940.

Conference Proceedings

Utley, J., Buzzard, G., Bouman, C., & Kemnetz, M. (2025). Boiling flow parameter estimation from boundary layer data. *SPIE Optics + Photonics*. DOI: 10.1117/12.3063655. Preprint: arXiv:2602.10394.

Utley, J., Buzzard, G., Bouman, C., & Kemnetz, M. (2024). Data-driven synthetic wavefront generation for boundary layer data. *SPIE Optics + Photonics*. DOI: 10.1117/12.3027740. Preprint: arXiv:2409.04873.

CONFERENCE PRESENTATIONS

Directed Energy S&T Symposium 2026 — "ReVAR: A data-driven algorithm for generating aero-optic phase screens."

Directed Energy Education Workshop 2026 — "ReVAR: A data-driven algorithm for generating aero-optic phase screens."

Electronic Imaging 2026 — "ReVAR: A data-driven algorithm for generating aero-optic phase screens."

SPIE Optics + Photonics 2025 — "Boiling flow parameter estimation from boundary layer data."

Directed Energy S&T Symposium 2025 — "Data-driven synthetic wavefront generation for boundary layer data."

Electronic Imaging 2025 — "Synthetic wavefront generation for aero-induced turbulence using boundary layer data."

SPIE Optics + Photonics 2024 — "Data-driven synthetic wavefront generation for boundary layer data."

Directed Energy S&T Symposium 2024 — "Synthetic wavefront generation for aero-optics correction."

HONORS, LEADERSHIP, SERVICE, & TEACHING

Fellowships: Doctoral Research Fellow, Oak Ridge Institute for Science and Education / AFIT (2025)

Awards: SPIE Student Conference Support Award (2024); Purdue College of Science Travel Award (2024); John H. Barrett Memorial Prize, University of Tennessee (2022).

Leadership: President, SIAM Student Chapter, Purdue University (2026–Present); Co-organizer, CCAM Lunch Seminar, Purdue Mathematics (2025–Present).

Service: Reviewer, *Optical Engineering* (SPIE), 2025–Present (2 manuscripts).

Teaching: Graduate TA, Purdue University (2022); Instructor, Wolverine Pathways at University of Michigan (2022–2023); Undergraduate TA & Math Tutor, University of Tennessee (2020–2022).